DiningMotion: Transforming Everyday Dining Objects into a Hand Movement Monitoring Platform

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Abstract

Everyday hand movements provide valuable information about fine motor function, especially in people with neurological disorders (e.g., Parkinson's). Monitoring these movements and tracking their changes in daily life are important for understanding long-term motor function, including rehabilitation monitoring, evaluation of treatment or tracking of disease progression. However, current systems often rely on body-worn sensors or specialized instrumented objects, which may require users to change their routines or be observed in experimental settings. Everyday dining utensils provide an opportunity to transform familiar objects into natural interfaces for hand motion monitoring. We present *DiningMotion*, a modular sensing system that transforms common dining objects into interfaces for movement sensing. *DiningMotion* uses a detachable wireless IMU and interchangeable clamp- and strap-based mounts to instrument everyday objects such as mugs, cups, and utensils. This low-cost, portable sensing module can be rapidly attached to different objects. The demonstration shows its ability to distinguish steady and tremor-like motion through real-time motion traces.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

Keywords

Fine motor monitoring, Object-mediated sensing, IMU sensing

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1 Introduction

Everyday hand movements valueble information about human mobility, functional performance, and fine-motor ability [3]. For people with neurological movement disorders, changes in hand movement can affect routine activities such as eating, drinking, and manipulating household objects [2]. Monitoring these movements in daily life is therefore important for long-term in-home health tracking, complementing periodic clinical assessments that only capture brief snapshots of motor function [1].

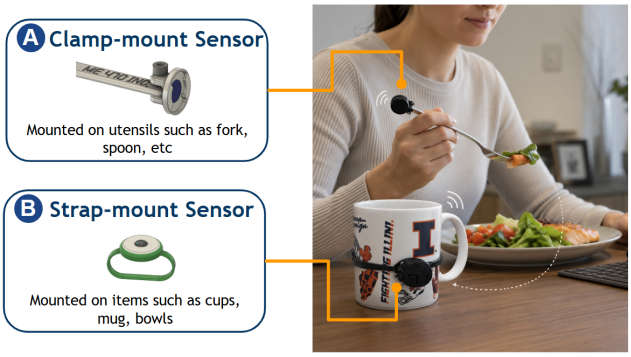


Figure 1: Concept of *DiningMotion*. Everyday dining objects, such as utensils and mugs, are instrumented with detachable sensors to capture natural hand-object interactions during in-home dining activities.

Existing motion monitoring systems often rely on body-worn sensors or specialized instrumented objects. These approaches often require users to change their routines or be observed under experiment condition [1]. Dining provides a natural sensing opportunity because it involves repeated interactions with familiar objects, including grasping, lifting, tilting, stabilizing, and utensil manipulation. We present *DiningMotion*, a modular sensing platform that transforms everyday dining objects into interfaces for movement sensing. As shown in Fig. 1, *DiningMotion* captures hand-object interactions during familiar dining activities rather than isolated hand motion. The system uses a detachable wireless IMU module and interchangeable mounts to instrument common objects such as mugs, cups, forks, and spoons. This demo shows how the same low-cost module can be rapidly attached to different dining objects, stream real-time motion traces, and distinguish steady and tremor-like object-mediated movements.

2 DiningMotion System

DiningMotion consists of three main components, as shown in Fig. 2. First, the sensing module integrates a wireless IMU, battery, and compact enclosure into a detachable unit. The module captures 6-DOF motion through acceleration and gyroscope measurements and can be reused across different dining objects. Second, *DiningMotion* uses interchangeable mounts to support different object geometries.

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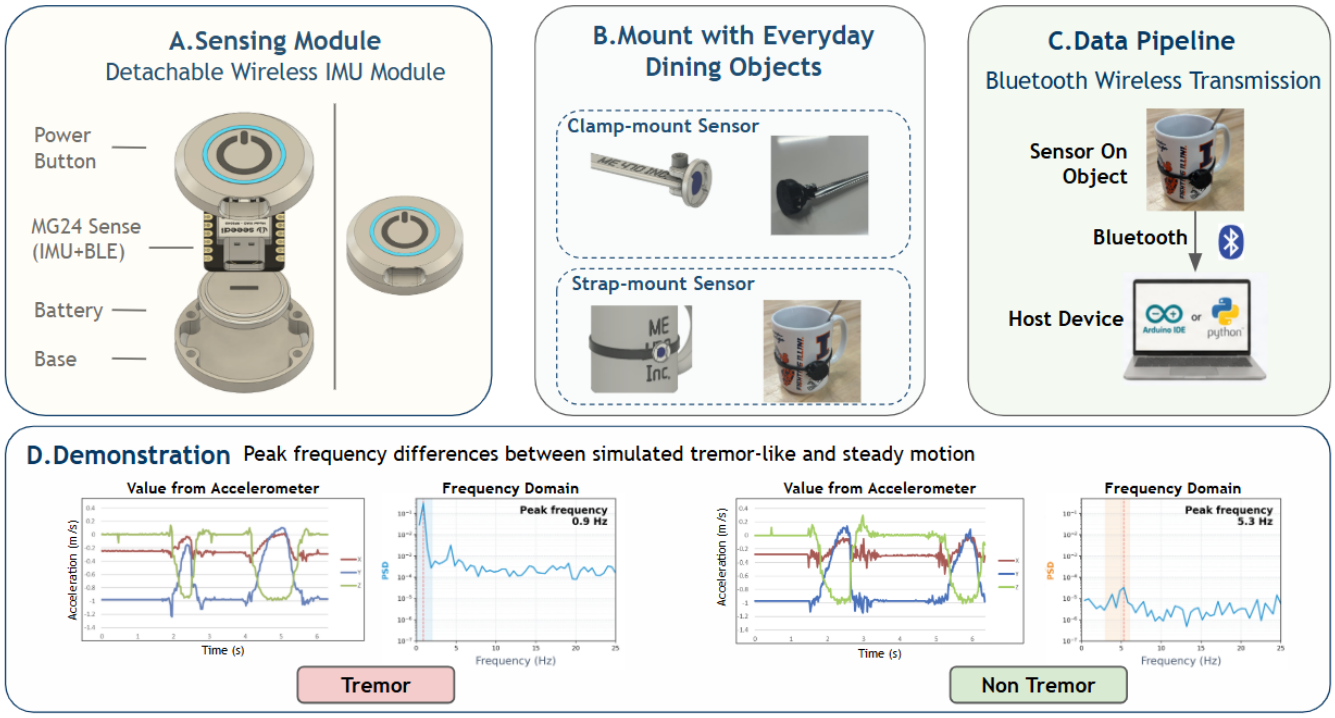


Figure 2: *DiningMotion* system overview. A detachable wireless IMU module attaches to everyday dining objects through interchangeable clamp- and strap-based mounts, streams motion data wirelessly, and supports real-time visualization and post-hoc analysis.

A clamp-based mount attaches to utensils such as forks and spoons, while a strap-based mount attaches to curved objects such as mugs and cups. This design allows everyday objects to be temporarily sensorized without replacing them with dedicated smart utensils.

Third, the instrumented object streams motion data to a host device through Bluetooth Low Energy. The host device receives timestamped acceleration and gyroscope data for live visualization and post-hoc processing. This pipeline supports real-time inspection of object-mediated hand motion as well as later analysis in the time and frequency domains.

Together, these components allow *DiningMotion* to transform common dining objects into sensing interfaces for everyday in-home movement monitoring.

3 Demonstration

Panel D in Fig. 2 shows example sensing signal and analytics output from the demonstration. The time-domain signal captures object motion during use, while the frequency-domain view reveals motion patterns associated with tremor-like movement. This demo focuses on system feasibility rather than clinical diagnosis, showing that the same low-cost module can be rapidly attached to different dining objects and capture distinguishable motion patterns during natural hand-object interaction.

4 Conclusion and Future Work

We presented *DiningMotion*, a modular sensing platform that transforms everyday dining objects into interfaces for in-home movement monitoring. By combining a detachable wireless IMU module, interchangeable mounts, and a Bluetooth data pipeline, *DiningMotion* enables mugs, cups, and utensils to capture object-mediated hand movements during familiar daily activities. The demonstration shows the feasibility of rapidly instrumenting different objects and visualizing steady and tremor-like motion patterns.

Future work will move beyond controlled demonstrations toward actual human-subject evaluation. We plan to refine the enclosure and operating procedure, conduct IRB-approved studies with participants, and evaluate *DiningMotion* for longitudinal tracking of dining-related hand movements in home environments.

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